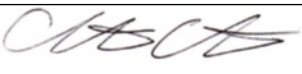


Guideline Title: Flight Physiology & Patient Oxygenation

Guideline Number: GUID-3203-G001

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** The effects of altitude on patients will be minimized during air transport. Patient oxygenation will be maintained during flight altitude.
- II. **DEFINITIONS:**
 - A. IV - Intravenous
- III. **DEPARTMENT GUIDELINE:**
 - A. All medical transport team members should be aware of the 8 Stressors of Flight. These not only affect the patient, but all members of the flight team as well.

i. Decreased partial pressure of oxygen	iv. Decreased humidity
ii. Decreased barometric pressure	v. Noise
iii. Thermal Changes	vi. Vibration
	vii. Fatigue
	viii. G-forces
 - B. All patients transported shall receive continuous pulse oximetry.
 - C. Supplemental oxygen shall be administered to all patients transported utilizing an appropriate delivery device and liter flow rate to maintain saturations > 92%.
 - D. Calculate the amount of oxygen required for every transport.
 - E. Infusion rates for intravenous fluids being infused via free-flow method may be affected by changes in altitude. All vasoactive medication infusions will be maintained via IV pumps. Free-flowing infusions will be monitored closely to ensure appropriate rates are maintained.
 - F. Invasive line considerations: Any air bubbles which exist within the intravenous/intra-arterial line systems will have increased volume at higher altitudes which could result in an increased probability of significant air embolus.
 - G. Do not occlude gastric tubes except for brief periods of time. Allow equilibration of intra-gastric pressures with prevalent atmospheric pressure.

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H. Along with routine monitoring of vital signs, all tubes and lines should be re-assessed every fifteen minutes at minimum.

I. GENERAL ASSESSMENT OF CHANGES ASSOCIATED WITH FLIGHT:

- i. Monitor patients for hypoxia.
- ii. Monitor patients for the following:
 - 1. Barotitis Media-trapped gas in the middle ear (ear pain).
 - 2. Barosinusitis-trapped gases in the sinuses (sinus pain).
 - 3. Barodentalgia-trapped gases in air pockets of the teeth (tooth pain).
 - 4. Gastrointestinal gas expansion.
 - 5. Pneumothoracies.
- iii. Monitor patients for air trapped in medical devices:
 - 1. Air in IV lines.
 - 2. Air in IV bags (causing fluids to run faster if not on an infusion pump).
 - 3. Air in endotracheal tube cuffs, balloon pump catheters, & pulmonary artery catheters.
- iv. Monitor patients for temperature changes and decreased humidity.
 - 1. Decreased humidity can cause mucous plugging, especially in neonates and pediatric patients who have small endotracheal tubes

IV. REFERENCES: None